Q1)

Code:

```
#!/bin/bash

read -p "Enter an IP address: " ip
if [[ $ip =~ ^([0-9]{1,3}\.){3}[0-9]{1,3}$ ]]; then
        valid=1
        oldIFS=$IFS
        IFS='.'
        for octet in $ip
        do
                if (( octet < 0 || octet > 255 )); then
                        echo "Ivalid IP: one or more octets are out of range (0-255)."
                        valid=0
                        break
                fi
        done
        IFS=$oldIFS

        if (( $valid )); then
                echo "Valid IP address."
        fi
else
        echo "Invalid IP Format"
fi
```

Output:

```
user@user:~/B24CS013$ ls
user@user:~/B24CS013$ vi B24CS013_01.sh
user@user:~/B24CS013$ ls
B24CS013_01.sh
user@user:~/B24CS013$ ls -l B24CS013_01.sh
-rw-rw-r-- 1 user user 569 Aug 28 16:41 B24CS013_01.sh
user@user:~/B24CS013$ chmod +x B24CS013_01.sh
user@user:~/B24CS013$ ls -l B24CS013_01.sh
-rwxrwxr-x 1 user user 569 Aug 28 16:41 B24CS013_01.sh
user@user:~/B24CS013$ ./B24CS013_01.sh
Enter an IP address: 4444.22.3.1
Invalid IP Format
user@user:~/B24CS013$ ./B24CS013_01.sh
Enter an IP address: 242.455.10.2
Ivalid IP: one or more octets are out of range (0-255).
user@user:~/B24CS013$ ./B24CS013_01.sh
Enter an IP address: 10.30.0.1
Valid IP address.
user@user:~/B24CS013$
```

Q2)

Code:

```
#!/bin/bash

max() {
        arr=("$@")
        max=${arr[0]}
        for i in "${arr[@]}"; do
                if ((i > max)); then
                        max=$i
                fi
        done
        echo $max
}

main (){
        read -p "Enter no.of elements: " n
        arr=()
        for ((i=0;i<n;i++)); do
                read num
                arr+=($num)
        done
        max=$(max "${arr[@]}")
        echo "Maximum element: $max"
}

main
```

Output:

```
user@user:~/B24CS013$ ls
B24CS013_01.sh
user@user:~/B24CS013$ vi B24CS013_02.sh
user@user:~/B24CS013$ ls -l B24CS013_02.sh
-rw-rw-r-- 1 user user 458 Aug 28 17:07 B24CS013_02.sh
user@user:~/B24CS013$ chmod +x B24CS013_02.sh
user@user:~/B24CS013$ ls -l B24CS013_02.sh
-rwxrwxr-x 1 user user 458 Aug 28 17:07 B24CS013_02.sh
user@user:~/B24CS013$ ./B24CS013_02.sh
Enter no.of elements: 3
1
3
7
Maximum element: 7
user@user:~/B24CS013$ ls
B24CS013_01.sh  B24CS013_02.sh
user@user:~/B24CS013$
```